GUNDLA MALLIKARJUN

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PROFESSIONAL SUMMARY

Results-driven **AI & Full Stack Software Engineer** with 2+ years of experience designing, building, and deploying production-grade **Generative AI, LLM, and cloud-native microservices** solutions at **Wabtec Corporation**. Proven track record of reducing diagnosis time by **40%** and incident triaging effort by **80%** through AI-driven automation, with hands-on experience across the full SDLC – from architecture and development to CI/CD and cloud deployment.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, C, C++, SQL
AI / ML: Generative AI, LLMs, RAG, LangChain, LangGraph, MCP, Semantic Search, Prompt Engineering, NLP, AWS Bedrock, Claude, GPT-4
Backend: Spring Boot, FastAPI, REST APIs, Microservices
Frontend: Angular, HTML5, CSS3, Streamlit
Databases: PostgreSQL, PgVector, SQL, Vector Databases
Cloud & DevOps: AWS (Bedrock, EC2, S3), Docker, Kubernetes (EKS), GitLab CI/CD, Linux
Security & Testing: JWT, OAuth 2.0, Pytest, Unit & Integration Testing, Secure API Design

PROFESSIONAL EXPERIENCE

Wabtec Corporation – AI Engineer July 2024 – Present

- Built a production-grade intelligent maintenance platform analyzing **500K+ locomotive maintenance records** using **FastAPI, PostgreSQL, PgVector, AWS Bedrock, and Claude**, cutting maintenance diagnosis time by over **40%**.
- Designed and deployed a custom **Model Context Protocol (MCP) server** exposing maintenance history, semantic search, and diagnostic intelligence to enterprise AI assistants through secure tool-driven interactions.
- Engineered a multi-stage maintenance issue classification pipeline combining exact matching, lemmatized token analysis, and **Claude Opus** inference, improving issue identification accuracy and consistency.
- Developed a hybrid semantic retrieval architecture using **Amazon Titan Embeddings**, **PgVector**, metadata filtering, fuzzy matching, and vector search to deliver explainable, context-aware recommendations.
- Integrated the **Amazon Bedrock Converse API** and implemented prompt optimization and response caching, reducing latency and inference costs while improving quality.
- Automated ingestion and processing of real-time work orders, enabling AI-assisted root cause analysis across the maintenance workflow.
- Provisioned and managed **Amazon EC2** instances as development environments, enabling consistent and reproducible team development workflows.

Wabtec Corporation – Full Stack Software Engineer Jan 2024 – June 2024

- Architected a centralized enterprise monitoring platform (**Angular, Spring Boot, PostgreSQL, Kafka, AWS**) covering **400+ Kafka topics, 90+ S3 pipelines, and 200+ microservices**, cutting manual incident triaging effort by **80%**.
- Automated CI/CD pipelines with **GitLab** and containerized services with **Docker** on **Amazon EKS**, improving release reliability and scalability.

PROJECTS

Multi-Agent Reinforcement Learning for CNN Hyperparameter Optimization

- Developed a **multi-agent reinforcement learning** system using **Q-learning** to automate CNN hyperparameter tuning.
- Designed a shared Q-table mechanism enabling collaborative learning across agents for faster convergence.
- Achieved **99.2% accuracy on MNIST** and **91% on Fashion-MNIST** through automated tuning.
- Implemented an **epsilon-greedy strategy** to balance exploration and exploitation during training.

FinTrack – Secure AI-Powered Personal Finance Platform

- Built a secure, full-stack AI-powered financial analytics platform using **Angular, Spring Boot, FastAPI, PostgreSQL, LangChain, and GPT-4.1**, covering document ingestion, transaction analysis, and dashboard visualization.
- Developed AI-powered transaction categorization and conversational insights using **LangChain** and Large Language Models, secured via **JWT authentication** and encrypted document handling.

EDUCATION

Master of Engineering – Computer Science and Information Security Aug 2022 – Jul 2024
National Institute of Technology Calicut CGPA: 8.75

Bachelor of Engineering – Computer Science Jul 2018 – May 2022
Osmania University CGPA: 7.95